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BFIN515 – Economic Analysis

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Assignment 2 – “Costco”

How does Costco's membership pricing affect membership demand, and what membership price would maximize estimated membership revenue?

For this research paper, I will be analyzing how Costco's membership pricing affects membership demand and what price specifically would maximize membership revenue. When it comes to decision making, the primary decision maker is "Costco Wholesale Corporation's Management, in which they're responsible for taking executive action on membership strategies and pricing.

1. Business Question

The goal for this research paper is to not only understand Costco's dataset and how they've adapted over the years to maximize their membership sales, but to see how correlation and effects of pricing and membership demand and what pricing specifically would maximize membership revenue. This would be of benefit to Costco Wholesale Corporations Management and the answer to this research question would be of use in determining the best pricing for members and can help form a greater understanding of their metrics for employees and executives.

Costco's business model operates as a membership-based warehouse retail model in which members pay an annual membership fee that grants them access to Costco's products and services. This is a significant source of recurring revenue for Costco and is prioritized at the front end. As a current employee of Costco, membership upgrades are strived for and are celebrated. For our business question, we are going to determine if Costco can increase its membership fee without seeing a reduction in membership demand. We will analyze the relationship between the annual membership fee and the number of Costco gold star cardholders and Executive cardholders. The last time we seen an increase in pricing for the Gold Star membership fee was from $60 to $65 back on September 1, 2024, as well as the Executive Membership from $120 to $130.

2. Data

With the data I collected, I was able to cover Costco's fiscal years 2014 through 2025 from the company's Form 10-K filings with the SEC. Each year included:

1. **Fees:** the U.S. and Canada Gold Star and Executive annual fees.
2. **Membership counts:** total paid members, Business members, Executive members, and total cardholders.
3. **Renewal rates:** the U.S./Canada and worldwide renewal rates.
4. **Revenue:** membership fee income.
5. **Warehouse count:** warehouses open at fiscal year-end, used as a control for expansion.

The annual fee for the gold star membership was $55 through the start of FY2017 and rose to $60 on June 1, 2017. It then rose to $65 on September 1, 2024. The Executive membership fee followed the same path, going from $110 to $120 to $130. The two most recent increases bracket a gap of roughly seven years which is longer than Costco's pattern of every five to six years.

Below is a table of my findings.

Variable	FY2016	FY2025
Gold Star fee	$55	$65
Total paid members (millions)	47.6	81.0
Total cardholders (millions)	86.7	145.2
Executive share of paid members	39%	47.8%
Renewal rate, U.S./Canada	90%	92.3%
Renewal rate, worldwide	88%	89.8%
Membership fee income ($ millions)	2,646	5,323
Fee income per paid member	$55.59	$65.72
Warehouses	715	914

In this data set we can see how paid memberships grew at a steady rate throughout the sample, as well as in the years with both fee increases. Renewal rates stayed above 90% in the U.S. and Canada. These patterns stay consistent with relatively inelastic demand, but they are also consistent with strong underlying growth that masks a price effect. The methods below are designed to separate the two. I've also noticed In Fiscal year 2020, Costco standardized its counting method globally, adding roughly 1.3 million paid members. This resulted in adjustments for the FY2020 growth rate. Another thing that I kept in mind was for Fiscal year 2014 thru Fiscal year 2019, Costco only reported a rounded Executive percentage of paid members (37%–39%).

3. Method

In my analysis I focused on studying the two fee increases, regression of membership growth on fee changes, and a revenue simulation that can estimate the elasticity to find the membership fee that maximizes on fee income. For **Revenue simulation,** Fiscal year 2025 membership fee income was scaled (R_0 = \$5,323 million) by the change in members and the change in average fee per member using the following formula:

$R(P) = R_0 \times [(1 - w) + w \times (Q(P)/Q_0) \times (P + c)/(P_0 + c)]$

The variables are defined as follows:

1. P is the Gold Star fee, and P_0 = \$65.
2. $c = s \times U = 0.478 \times \$65 = \$31.06$, the average Executive upgrade revenue per member that moves with P. Here s is the Executive share of paid members and U is the Executive upgrade fee.
3. $w = 52/76.2 = 68.2\%$, the share of members subject to the U.S./Canada fee.
4. Q(P) is membership demand at fee P.

   I compare three demand shapes, each calibrated to the elasticity ε from Step 1 (the average of the two event-study estimates, −0.34):
5. **Model A (constant elasticity):** $Q = Q_0(P/P_0)^{\varepsilon}$. If $|\varepsilon| < 1$, revenue rises with price at every level, so this model has no interior maximum.
6. **Model B (linear demand):** $Q = Q_0 + (\varepsilon Q_0/P_0)(P - P_0)$. Revenue is maximized at $P^* = -[P_0(1 - \varepsilon) + \varepsilon c] / (2\varepsilon)$.

7. **Model C (rising elasticity):** $Q = Q_0 \cdot \exp[(\varepsilon/k)((P/P_0)^k - 1)]$. Demand becomes more price-sensitive at higher fees.

4. Results

Paid membership grew after both fee increases, but in a slower gradual pace. Elasticity was −0.52 after the 2017 increase (+9.1%) and −0.16 after the 2024 increase (+8.3%), averaging −0.34. The growth regressions give a similar range: −0.14 in Model 1 and −0.38 cumulative in Model 2. Overall, demand appears to be inelastic in the $55–$65 range.

Applying the average elasticity of −0.34 to Fiscal Year 2025 fee income of $5,323 million gives these revenue-maximizing Gold Star fees:

Demand model	**Revenue-maximizing fee**	**Revenue at maximum**
Linear demand	about $114	$5,785M (+8.7%)
Rising elasticity (k = 2, assumed)	$100	$5,629M (+5.7%)
Constant elasticity	No interior maximum	n/a

Revenue is flat near the peak. A $70 fee raises estimated income by only about $90 million (+1.7%), while the optimum falls to about $82 at an elasticity of −0.5.

5. Validation

In terms of Validation, when comparing to Costco's own figures, their Fiscal Year 2025 attributes about 40% of that year's fee income growth to the fee increase. 52 million affected memberships at an average increase of about $7.50 which is about $390 million a year, which brings closely to the company's $370 million two-year estimate. Costco also attributes part of the FY2025 renewal-rate dip to online sign-ups, which renew at lower rates. This means the 2024 shortfall overstates the price effect.

6. Recommendation

One recommendation I have would be that Costco's management should continue raising membership fees in small steps rather than a large one-time jump. Demand is inelastic so each step raises revenue. A $5 increase to $70 would add about $90 million in annual fee income under the linear demand model. I feel Management should also watch the executive tier closely and how Executive members now make up 47.8% of paid members, and their 2% reward cost rises with the fee, which narrows the net gain from further increases. I also think that Costco should continue the gap of five to six years between increases since data shows it worked out well rather than breaking it again and doing two increases.

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